# Functional Annotation Report: RTL8B (Retrotransposon Gag-like protein 8B)

**Gene:** RTL8B (synonyms: CXX1C, FAM127C, MAR8B / MART8B) **UniProt:** Q17RB0 · **Organism:** *Homo sapiens* · **Locus:** X chromosome (Xq23) **Family:** FAM127 / Mart / SIRH (Sushi-ichi-related retrotransposon homolog) family **Domain:** RTL1/LDOC capsid-like (InterPro IPR032549; Pfam PF16297) **Date:** 2026-10-24

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## 1. Summary (Answer to the Research Question)

RTL8B is a small (~113 aa, ~13 kDa) **domesticated retrotransposon Gag/capsid-derived protein**. It is not an enzyme, transporter, or classical signaling molecule; rather it is a **capsid-homology adaptor/modulator** consisting essentially of a single N-terminal Gag "capsid-like" domain. Its established primary function is **protein–protein interaction with two other retroelement-derived human proteins — the Gag-like protein PEG10 (RTL2) and the ubiquitin-proteasome shuttle factor UBQLN2 — to control their subcellular localization and activity**. Specifically, RTL8(B):

1. **binds the N-terminal capsid domain of PEG10** and acts as a **dominant-negative "capsid poison,"** reducing assembly/release of PEG10 virus-like particles (VLPs) and retaining PEG10 intracellularly;
2. **promotes nuclear/nucleolar translocation of UBQLN2** into ubiquitin-rich sub-nuclear protein-quality-control (PQC) compartments; and
3. is the **obligatory bridging factor** that enables PEG10 to recruit UBQLN2 into cytoplasmic **stress granules**, thereby shaping stress-granule dynamics and proteostasis.

Thus RTL8B operates at the **intersection of endogenous retroelement capsid biology and cellular protein quality control**, acting in both the **nucleolus/nucleus** and the **cytoplasm (stress granules)**.

> **Identity caveat (addressed):** "RTL8B" (Q17RB0) is unambiguously the human FAM127/Mart Gag-like gene described below. The published functional literature refers to the protein collectively as **"RTL8"** because humans carry three X-linked paralogs — RTL8A,

> RTL8B, RTL8C — encoding **near-identical** proteins. Sequence comparison performed in this analysis shows RTL8B is **96.5% identical (109/113 residues)** to RTL8A (Q9BWD3) and nearly identical to RTL8C (A6ZKI3). The paralogs cannot be distinguished by the antibodies or knockdown/overexpression reagents used in the primary studies, so the functional conclusions apply to RTL8B.

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## 2. Molecular Identity and Structure

- **Protein:** 113 residues, ~13 kDa; sequence (Q17RB0): `MEGRVQLMKALLARPLRPAARRWRNPIPFPETFDGDTDRLPEFIVQTSSYMFVDENTFSNDALKVTFLITRLTGPALQWVIPYIKKESPLLSDY`
- **Domain architecture:** a single **RTL1/LDOC capsid-like domain** (PF16297) spanning essentially the whole protein. It corresponds to the **N-terminal Gag capsid (CA) region of a Sushi-ichi/LTR retrotransposon**, and resembles a **truncated PEG10** that retains only the capsid-homology portion — it **lacks any Pol/protease, reverse-transcriptase, or nucleocapsid (zinc-knuckle) region** (Campodonico et al. 2024, P 39775359).
- **Paralog set (this analysis):** RTL8B vs RTL8A = 96.5% identity; RTL8C near-identical. All three are small single-exon-like Gag capsid genes clustered on the X chromosome.
- **Family placement:** member of the **SIRH/Mart/RTL family** of genes domesticated from LTR retrotransposons in mammals; related members include PEG10 (SIRH1/RTL2), RTL1/PEG11 (SIRH2), and LDOC1 (RTL7/SIRH7) (Kaneko-Ishino & Ishino 2012, P 22866050).

**Functional inference from structure/evolution:** Possessing only a capsid-homology domain and no catalytic module, RTL8B is predicted to act through **homotypic/heterotypic capsid–capsid interactions** rather than catalysis — consistent with all experimental data showing it works by binding and modulating partner proteins.

**Bioinformatic corroboration (this analysis):** A BLOSUM62 Smith-Waterman alignment of RTL8B against PEG10 shows RTL8B residues **26–105 align to PEG10 residues ~79–158** at **35% identity / 54% similarity** (80-aa window), with conserved capsid motifs (…PE·FDG…, and a C-terminal …**MK·VF** block: RTL8B `AEMKRVF` vs PEG10 `MMEMKHVF`). The alignment covers ~70% of RTL8B — i.e. essentially the **entire protein is one copy of PEG10's N-terminal Gag capsid**

**domain** — and the homologous stretch **coincides with the experimentally-mapped RTL8-binding region of PEG10 capsid** ( ).

P 39775359

This independently supports a capsid–capsid heterodimerization mechanism.

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# 3. Primary Function and Mechanism

## 3.1 Antagonist of PEG10 capsid / virus-like-particle assembly

PEG10 is a Gag-Pol–like, retroelement-derived protein that **self-assembles into virus-like particles (VLPs)** which package mRNA and are secreted in extracellular vesicles (Segel et al. 2021, ).

P 34413232

RTL8 shares homology with the PEG10 **N-terminal capsid domain** and binds there. On its own RTL8 **cannot form VLPs**, but it is **incorporated into PEG10 particles**; when co-expressed, RTL8 **decreases PEG10 VLP abundance and raises intracellular PEG10**, i.e., it **inhibits PEG10 VLP formation/release** (Campodonico et al. 2024, ).

P 39775359

> **Mechanistic model:** RTL8B behaves like a **capsid "chain terminator"/dominant-negative subunit** — it co-assembles with PEG10 capsid via the shared Gag interface but, lacking the full assembly determinants, caps or poisons higher-order particle formation, thereby limiting VLP output and retaining PEG10 in the cell.

## 3.2 Adaptor controlling UBQLN2 localization (nuclear PQC)

RTL8 was discovered as a **UBQLN2 interactor** by mass spectrometry and validated with recombinant proteins and in mouse brain (Mohan et al. 2022, ).

P 35247097

Endogenous and overexpressed **RTL8 localizes to nucleoli**; co-expressed RTL8 **promotes nuclear translocation of UBQLN2**, including under heat-shock stress, and the two **colocalize**

**in ubiquitin-enriched sub-nuclear structures containing protein-quality-control components**. The effect is **selective for UBQLN2** (not UBQLN1 or UBQLN4), and UBQLN2 reciprocally stabilizes RTL8 protein levels.

### 3.3 Obligatory bridge in the UBQLN2–PEG10 stress-granule axis

RTL8 **"confers on UBQLN2 the ability to complex with and regulate PEG10."** PEG10 is a core stress-granule component and recruits UBQLN2 to stress granules under stress, **but only when RTL8 is present**. Perturbing RTL8 (or UBQLN2/PEG10) levels **remodels stress-granule disassembly kinetics and translation recovery**; within stress granules PEG10 forms VLPs (Mohan et al. 2025, P 40680123; preprint P 39484508).

**Synthesis of mechanism:** RTL8B is a **molecular linchpin** that uses its single capsid interface to simultaneously engage PEG10 (another capsid protein) and the UBQLN2 proteostasis shuttle, thereby (i) tuning PEG10 capsid/VLP assembly and (ii) routing UBQLN2 between cytoplasmic stress granules and nuclear/nucleolar PQC compartments.

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## 4. Subcellular Localization

| Compartment | Evidence |
|---|---|
| **Nucleolus** (steady state) | RTL8 localizes to nucleoli under endogenous and overexpressed conditions ( P 35247097 ); rationalized by an N-terminal arginine-rich NoLS-like motif (see below) |
| **Nucleus / sub-nuclear PQC foci** | RTL8 + UBQLN2 colocalize in ubiquitin-enriched sub-nuclear structures ( P 35247097 ) |
| **Cytoplasm / stress granules** | RTL8 required for PEG10-mediated UBQLN2 recruitment to stress granules ( P 40680123 ) |
| **Incorporated into PEG10 VLPs** | RTL8 packaged into PEG10-derived particles ( P 39775359 ) |

RTL8B therefore acts in **multiple compartments**, shuttling/partitioning between the nucleolus/nucleus and cytoplasmic condensates depending on its partners and cellular stress state.

**Sequence rationale for nucleolar targeting (this analysis):** RTL8B's N-terminus is strongly basic — **6 arginines in the first 25 residues**, with a maximal positive-charge window at residues ~13–26 (`RPLRPAARRWRN`, net +4), conserved across RTL8A/B/C and lying just upstream of the capsid-homology region. This architecture matches a canonical **arginine-rich nucleolar-localization signal (NoLS)/nucleic-acid-binding motif**, providing a mechanistic explanation for RTL8B's steady-state nucleolar localization. Notably, the PEG10 N-terminus lacks a comparable arginine cluster, consistent with PEG10 not being intrinsically nucleolar. (Bioinformatic inference; not yet confirmed by mutagenesis.)

## 5. Pathways and Biological Context

- **Endogenous retroelement capsid pathway:** regulator of PEG10 VLP biogenesis (P 39775359). P 34413232).
- **Protein quality control / ubiquitin-proteasome system (proteostasis):** governs localization of the UBQLN2 shuttle factor and the kinetics of stress-granule assembly/ disassembly (P 35247097). P 40680123).
- **Tissue/expression context:** RTL8 is broadly expressed and co-expressed with PEG10 in many tissues including **human brain** (P 39775359). Because UBQLN2 and PEG10 are implicated in **ALS/FTD and other neurodegenerative disease**, RTL8B's control of their localization and assembly is of likely neurobiological relevance.
- **Evolutionary pathway:** domesticated from a **Sushi-ichi/LTR retrotransposon Gag** during mammalian evolution; part of the SIRH/Mart family whose founding members (PEG10, RTL1) are essential for placentation, situating RTL8 within a retroelement-derived regulatory network (P 22866050).

## 6. Supported and Refuted Hypotheses

**Supported (experimental evidence):** - H1 — RTL8B is a capsid-homology adaptor, not an enzyme. ✔ (domain structure; truncated-Gag homology, P 39775359)
- H2 — RTL8 directs UBQLN2 to the nucleus/nucleolus and PQC foci. ✔

( P 35247097 )

- H3 — RTL8 binds PEG10 capsid and antagonizes VLP formation. ✔ ( P 39775359 )

- H4 — RTL8 is required to bridge PEG10 and UBQLN2 at stress granules. ✔ ( P 40680123 )

**Refuted / unsupported:** - RTL8B is NOT an active retrotransposon, reverse transcriptase, or protease — it retains no Pol/catalytic domain. - RTL8B does NOT act on UBQLN1 or UBQLN4 — the effect is UBQLN2-selective ( P 35247097 ).

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## 7. Limitations and Future Directions

- **Paralog resolution:** reagents do not distinguish RTL8A/B/C; the specific, non-redundant contribution of **RTL8B** versus its paralogs has not been isolated. Paralog-specific CRISPR/degron studies are needed. This near-identity (~96–97%) also defeats high-throughput resources: in this analysis **GTEx v8 returned no median expression** for RTL8B/A/C and the **AlphaFold DB has no deposited model for Q17RB0** (HTTP 404) — multi-mapping reads and redundant sequences prevent unique assignment. Expression context therefore rests on the primary-literature statement that RTL8 is broadly expressed, including in human brain ( P 39775359 ).
- **In-vivo physiology:** no dedicated RTL8B knockout phenotype is reported; organismal function is inferred from in-vitro/cell and mouse-brain interaction data.
- **Structure:** no experimental 3D structure of RTL8B–PEG10 or RTL8B–UBQLN2 complexes; the predicted capsid fold and binding interface merit structural confirmation (cryo-EM/AlphaFold-Multimer).
- **Directionality:** whether RTL8B's nucleolar role and its cytoplasmic stress-granule role are competing or sequential activities remains open.

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## 8. Key References


- Mohan et al. 2022, *J Biol Chem/Hum Mol Genet* — RTL8 promotes nuclear localization of UBQLN2 ( PMID 35247097 ).
- Campodonico et al. 2024, *(Whiteley/Sharkey labs)* — RTL8 antagonizes PEG10 VLPs ( PMID 39775359 ).
- Mohan et al. 2025 — Retrovirus-like proteins recruit UBQLN2 to stress granules ( PMID 40680123 ; preprint 39484508).
- Segel et al. 2021, *Science* — PEG10 packages its own mRNA / VLPs ( PMID 34413232 ).
- Kaneko-Ishino & Ishino 2012 — Domesticated LTR-retrotransposon genes in mammals ( PMID 22866050 ).